

A class data question

Class activity: investigate the number of files students downloaded yesterday.

What exactly should count as one download?

Collecting observations

Use anonymous IDs and record one download count per willing participant.

What recording rules would make everyone's counts comparable?

A systematic sample

Practice roster: IDs 01-24. Select 8 IDs using a random start from 1, 2, or 3 and a step of 3.

Which IDs does our random start select?

A practice dataset

Synthetic counts for 24 students, in ID order:

0, 2, 4, 1, 3, 2, 5, 0, 1, 2, 6, 3,
1, 4, 2, 0, 5, 3, 2, 1, 4, 2, 3, 6

What are the download counts for our 8 selected IDs?

An exact-value table

Use our 8 selected download counts.

What frequency table can we make for the values 0 through 6?

Relative frequencies

Use our exact-value table for the 8 sampled students.

What are the relative frequencies?

Cumulative frequencies

Use our exact-value table for the 8 sampled students.

What are the cumulative relative frequencies?

A grouped table

Regroup the same sample into 0-1, 2-3, 4-5, and 6 or more downloads.

What relative frequency table results?

Information lost in grouping

Use our exact-value and grouped tables.

Which table can determine the exact percentage with at most 2 downloads?

Interpreting a cutoff

Use our exact-value and grouped tables.

Can both tables determine the exact percentage with at most 3 downloads?

Above a cutoff

Use the exact-value and grouped frequency tables from our download-count activity.

Which table determines the exact percentage with more than 2 downloads?

Above a class boundary

Use the exact-value and grouped frequency tables from our download-count activity.

Can both tables determine the percentage with more than 3 downloads?

Choosing a grouping

The same observations can be grouped as 0–1, 2–3, 4–5, 6+, or displayed separately.

Is one table more correct, or do the tables answer different questions?

A different starting point

Repeat the systematic sample with a different starting ID.

How does the percentage with at most 2 downloads change?

Scope of the evidence

A class collects real download counts only from its own willing participants.

Which population could this sample reasonably describe?